

Airport Operations & Passenger Traffic Analysis - SQL Project

Top Airlines by Passenger Traffic

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view containing 'airport_analysis', 'airport_project_data', 'Views', 'Stored Procedures', 'Functions', 'sakila', 'sys', and 'world'. The main query editor shows the following SQL query:

```
1 SELECT
2   UNIQUE_CARRIER_NAME,
3   SUM(PASSENGERS) AS Total_Passengers
4 FROM Airport_Project_Data
5 GROUP BY UNIQUE_CARRIER_NAME
6 ORDER BY Total_Passengers DESC
7 LIMIT 10;
```

The 'Result Grid' at the bottom displays the results of the query:

UNIQUE_CARRIER_NAME	Total_Passengers
Southwest Airlines Co.	10998054
Delta Air Lines Inc.	9739964
American Airlines Inc.	9548187
United Air Lines Inc.	8089034
SkyWest Airlines Inc.	3196083
Spirit Air Lines	2533116
Alaska Airlines Inc.	2326633
Frontier Airlines Inc.	2287716
JetBlue Airways	2144165
Republic Airline	1419088

The 'Output' panel at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	10:32:13	USE airport_analysis	0 row(s) affected	0.000 sec
2	10:32:18	DESCRIBE Airport_Project_Data	36 row(s) returned	0.000 sec / 0.000 sec
3	10:32:59	SELECT UNIQUE_CARRIER_NAME, SUM(PASSENGERS) AS Total_Passengers F...	10 row(s) returned	0.110 sec / 0.000 sec

Top Origin Airports by Passenger Traffic

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

airport_analysis

Tables

airport_project_data

Views

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Functions

sys

world

Query 1

```
1 SELECT
2 ORIGIN,
3 SUM(PASSENGERS) AS Total_Passengers
4 FROM Airport_Project_Data
5 GROUP BY ORIGIN
6 ORDER BY Total_Passengers DESC
7 LIMIT 10;
```

Result Grid

ORIGIN	Total_Passengers
ATL	3095254
DEN	2728395
DFW	2575266
ORD	2139178
MCO	2001905
LAS	1969324
PHX	1811982
CLT	1793744
LAX	1774710
SEA	1502472

Administration Schemas Information

No object selected

Result 3

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:32:13	USE airport_analysis	0 row(s) affected	0.000 sec
2	10:32:18	DESCRIBE Airport_Project_Data	36 row(s) returned	0.000 sec / 0.000 sec
3	10:32:59	SELECT UNIQUE_CARRIER_NAME, SUM(PASSENGERS) AS Total_Passengers F...	10 row(s) returned	0.110 sec / 0.000 sec
4	10:34:09	SELECT ORIGIN, SUM(PASSENGERS) AS Total_Passengers FROM Airport_Projec...	10 row(s) returned	0.031 sec / 0.000 sec

Object Info Session

Top Origin Cities by Passenger Traffic

MySQL Workbench

Local instance MySQL80

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Stored Procedures

Functions

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world

Query 1

```
1 SELECT
2 ORIGIN_CITY_NAME,
3 SUM(PASSENGERS) AS Total_Passengers
4 FROM Airport_Project_Data
5 GROUP BY ORIGIN_CITY_NAME
6 ORDER BY Total_Passengers DESC
7 LIMIT 10;
```

Result Grid

ORIGIN_CITY_NAME	Total_Passengers
Atlanta, GA	3095302
Denver, CO	2728401
Chicago, IL	2678550
Dallas/Fort Worth, TX	2576274
New York, NY	2133017
Orlando, FL	2001919
Las Vegas, NV	1969356
Phoenix, AZ	1901988
Charlotte, NC	1793744
Los Angeles, CA	1774710

Administration Schemas Information

No object selected

Result 4

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:32:13	USE airport_analysis	0 row(s) affected	0.000 sec
2	10:32:18	DESCRIBE Airport_Project_Data	36 row(s) returned	0.000 sec / 0.000 sec
3	10:32:59	SELECT UNIQUE_CARRIER_NAME, SUM(PASSENGERS) AS Total_Passengers F...	10 row(s) returned	0.110 sec / 0.000 sec
4	10:34:09	SELECT ORIGIN, SUM(PASSENGERS) AS Total_Passengers FROM Airport_Projec...	10 row(s) returned	0.031 sec / 0.000 sec
5	10:34:59	SELECT ORIGIN_CITY_NAME, SUM(PASSENGERS) AS Total_Passengers FROM ...	10 row(s) returned	0.062 sec / 0.000 sec

Object Info Session

Passenger Traffic by State

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

airport_analysis

Tables

airport_project_data

Views

Stored Procedures

Functions

sakila

sys

world

Query 1

```

1 • SELECT
2   ORIGIN_STATE_ABR,
3   SUM(PASSENGERS) AS Total_Passengers
4 FROM Airport_Project_Data
5 GROUP BY ORIGIN_STATE_ABR
6 ORDER BY Total_Passengers DESC;

```

Result Grid

ORIGIN_STATE_ABR	Total_Passengers
FL	7427010
CA	6342745
TX	6292505
GA	3242322
CO	3035532
NY	2789638
IL	2776136
NC	2493422
NV	2148048
AZ	2070889
VA	1795589
WA	1751189
HI	1374605
NJ	1335703
PA	1270041
MI	1224031
MN	1136777
MA	1081375
UT	1052909
TN	1042848
MO	874454
MD	748713
OR	701621
OH	697064
PR	592172
LA	526623
KY	466127
IN	422153
SC	387874

Output

#	Time	Action	Message	Duration / Fetch
1	10:32:13	USE airport_analysis	0 row(s) affected	0.000 sec
2	10:32:18	DESCRIBE Airport_Project_Data	36 row(s) returned	0.000 sec / 0.000 sec
3	10:32:59	SELECT UNIQUE_CARRIER_NAME, SUM(PASSENGERS) AS Total_Passengers...	10 row(s) returned	0.110 sec / 0.000 sec
4	10:34:09	SELECT ORIGIN_STATE_ABR, SUM(PASSENGERS) AS Total_Passengers FROM Airport_P...	10 row(s) returned	0.031 sec / 0.000 sec

Average Route Distance Analysis

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

airport_analysis

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airport_project_data

Views

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Functions

sakila

sys

world

Query 1

```

1 • SELECT
2   AVG(DISTANCE) AS Average_Distance
3 FROM Airport_Project_Data;

```

Result Grid

Average_Distance
804,6739435912878

Output

#	Time	Action	Message	Duration / Fetch
5	10:34:59	SELECT ORIGIN_CITY_NAME, SUM(PASSENGERS) AS Total_Passengers FRO...	10 row(s) returned	0.062 sec / 0.000 sec
6	10:52:31	SELECT ORIGIN_STATE_ABR, SUM(PASSENGERS) AS Total_Passengers FRO...	53 row(s) returned	0.031 sec / 0.000 sec
7	10:53:28	SELECT AVG(DISTANCE) AS Average_Distance FROM Airport_Project_Data LIM...	1 row(s) returned	0.015 sec / 0.000 sec

Top Airlines by Freight Volume

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

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SCHEMAS

Filter objects

airport_analysis

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airport_project_data

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sys

world

Query 1

```
2 UNIQUE_CARRIER_NAME,  
3 SUM(FREIGHT) AS Total_Freight  
4 FROM Airport_Project_Data  
5 GROUP BY UNIQUE_CARRIER_NAME  
6 ORDER BY Total_Freight DESC  
7 LIMIT 10;
```

Result Grid

UNIQUE_CARRIER_NAME	Total_Freight
Federal Express Corporation	618065557
United Parcel Service	561875912
Air Transport International	154763076
Atlas Air Inc.	143094649
ABX Air Inc.	96458682
Kalitta Air LLC	57646331
Hawaiian Airlines Inc.	31414787
United Air Lines Inc.	30835083
Delta Air Lines Inc.	22902763
Sun Country Airlines d/b/a MN Airlines	22651176

Administration Schemas

Information

No object selected

Result 7

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	10:52:31	SELECT ORIGIN_STATE_ABR, SUM(PASSENGERS) AS Total_Passengers FROM...	53 row(s) returned	0.031 sec / 0.000 sec
7	10:53:28	SELECT AVG(DISTANCE) AS Average_Distance FROM Airport_Project_Data LIM...	1 row(s) returned	0.015 sec / 0.000 sec
8	10:53:47	SELECT UNIQUE_CARRIER_NAME, SUM(FREIGHT) AS Total_Freight FROM Air...	10 row(s) returned	0.062 sec / 0.000 sec

Object Info Session

Passenger Traffic by Region

MySQL Workbench

Local instance MySQL80

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Navigator

SCHEMAS

Filter objects

airport_analysis

Tables

airport_project_data

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Functions

sakila

sys

world

Query 1

```
1 SELECT  
2 REGION,  
3 SUM(PASSENGERS) AS Total_Passengers  
4 FROM Airport_Project_Data  
5 GROUP BY REGION  
6 ORDER BY Total_Passengers DESC;
```

Result Grid

REGION	Total_Passengers
D	60584340
L	11368
P	4621
I	3062
A	0

Administration Schemas

Information

No object selected

Result 8

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	10:53:28	SELECT AVG(DISTANCE) AS Average_Distance FROM Airport_Project_Data LIM...	1 row(s) returned	0.015 sec / 0.000 sec
8	10:53:47	SELECT UNIQUE_CARRIER_NAME, SUM(FREIGHT) AS Total_Freight FROM Air...	10 row(s) returned	0.062 sec / 0.000 sec
9	10:56:25	SELECT REGION, SUM(PASSENGERS) AS Total_Passengers FROM Airport_Proj...	5 row(s) returned	0.031 sec / 0.000 sec

Object Info Session